

CNS

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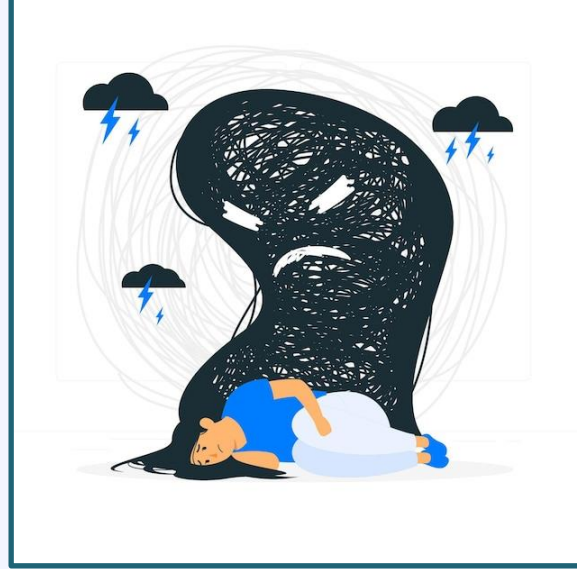
05 Depression





Introduction

- Depression is a disorder of mood.
- Globally it affects over 350 million people all over the world.
- It affects women more than men.
- Depression is considered a major cause of disability.
- Depression can lead to suicide.



Pathophysiology

Genetics:

- More than 4 genes were identified to cause depression.



- Biogenic amines and receptor theory:
 - Low levels of dopamine, serotonin, nor-epinephrine in brain.
 - Upregulation of certain receptors in the brain eg: 5HT2a and 5HT2c
 - This theory is the most accepted theory.
- Note: bio amine theory and genetics offer a short-handed explanation when it comes to trauma or experience caused depression.

Signs & Symptoms

- Symptoms of Depression:

- Emotional Symptoms :

- low levels of serotonin are the cause of emotional symptoms.
- loss of interest and enjoyment.
- Apathy. (lack of interest and emotional expression)
- loss of Self steam.
- Loss of Motivation.

- Biological Symptoms (Psycho-somatic disorder):

- low levels of nor-epinephrine are the cause of biological symptoms.
- Chronic pain.
- Loss of appetite.
- Sleep disorder.

BACKGROUND

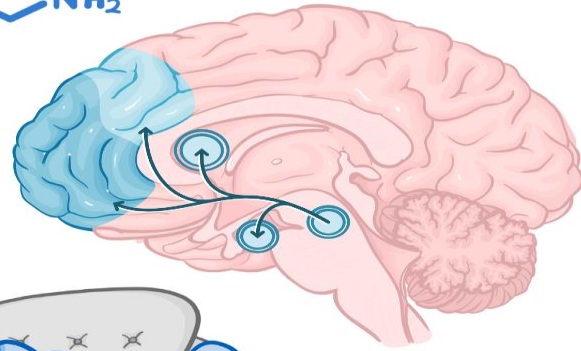
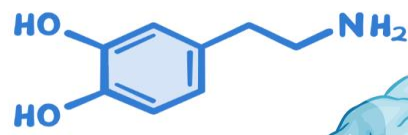
* LOSS of ABILITY to FEEL PLEASURE

~ COMMON SYMPTOM of DEPRESSIVE & SUBSTANCE USE DISORDERS



DIAGNOSIS

- * SNAITH-HAMILTON PLEASURE SCALE (SHAPS)
- * POSITIVE & NEGATIVE AFFECT SCHEDULE (PANAS)
- * ADDITIONAL TESTS
 - ~ SUBSTANCE USE
 - ~ VITAMIN DEFICIENCIES
 - ~ THYROID FUNCTION



CAUSES

- * DEPRESSIVE DISORDERS
- * TRAUMA or STRESS
- * DECREASED ACTIVITY in VENTRAL STRIATUM
- * DYSREGULATION of HORMONES
 - ~ GABA, DOPAMINE



TREATMENT

- * COGNITIVE BEHAVIORAL THERAPY
- * AEROBIC EXERCISE
- * ELECTROCONVULSIVE THERAPY
- * MEDICATION



Anti-depressant drugs

- Families are:
 - Monoamine oxidase inhibitors (MAO). 1960
 - Tricyclic anti-depressants (TCA). 1970
 - Selective serotonin reuptake inhibitors (SSRI). 1990
 - Atypical anti-depressants (heterocyclic). 2000
- SSRI and TCA are the most used drugs in treatment of depression.
- Anti-depressants need 2 weeks to produce a clinically significant effect, maximum benefit is observed at 12 weeks.



Medication and psychotherapy

- Despite their obvious clinical efficacy in treatment of mood disorder associated with depression and other diseases like general anxiety disorder (GAD), anti-depressants are not the first line treatment. **Psychotherapy with or without medication is the first line treatment to depression.**
- There is a great debate within the scientific community about when should anti-depressants be prescribed as only using anti-depressants without psychotherapy has caused cases of rebound depression once drug therapy ceases even when proper guidelines are followed.



Monoamine Oxidase inhibitors (MAO inhibitors)

Monoamine oxidase enzyme has to 2 sub types:

- Type A present in brain and GIT
- Type B in brain only and metabolizes dopamine only

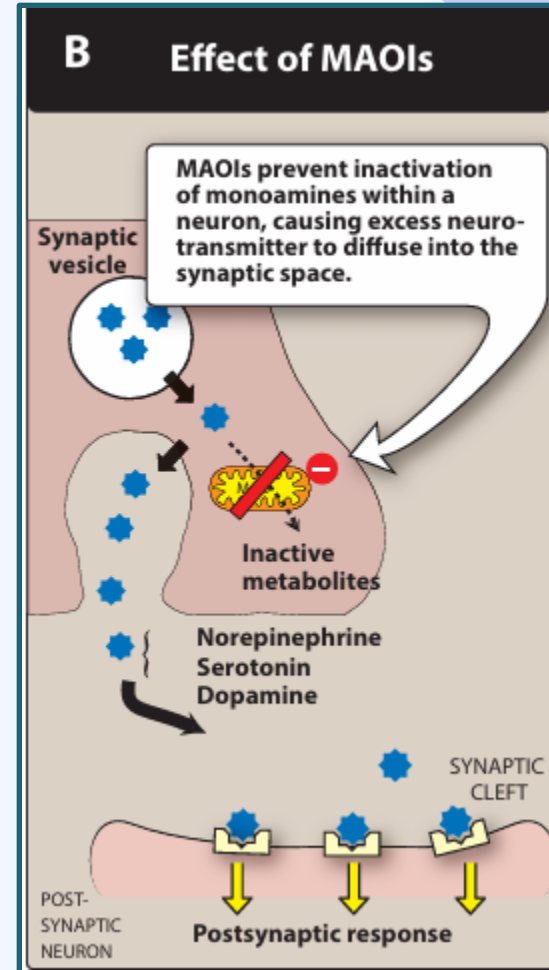
- Members:

- Phenelzine (A&B)
- tranylcypromine (A&b)
- selegiline. (B only)

- M.O.A:

- They prevent metabolism of reuptaken bio amines by MAO enzyme, thus increasing the amount available of bio amines to the receptors.

- MAO inhibitors are **not used as first line** anti-depressant drug due to their Severe side effects, but used for People who are unresponsive to other antidepressant drugs.



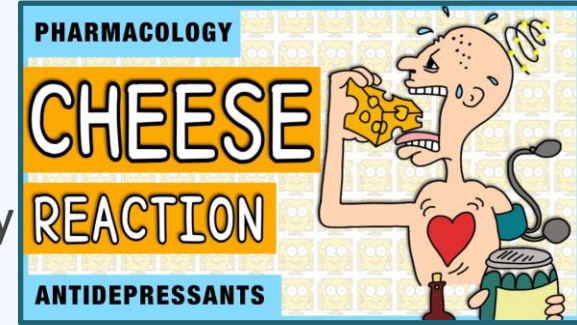
- Adverse effects:

- The following adverse effects are common with all anti-depressants:

- Orthostatic (Postural) hypotension, due to overall increase of bio amines in brain.
 - Sexual dysfunction in the form of delayed orgasm. This effect can treat pre-mature ejaculation, however, if the problem is not present this drug causes dysfunction.
 - Serotonin Syndrome (hyperthermia , arrhythmia and increased mortality rate) this effect increases significantly when combined with other serotonergic drugs.
 - CNS abnormalities depending on the drug
 - stimulation (Irritability and convulsions)
 - Sedation. (sleep and drowsiness)
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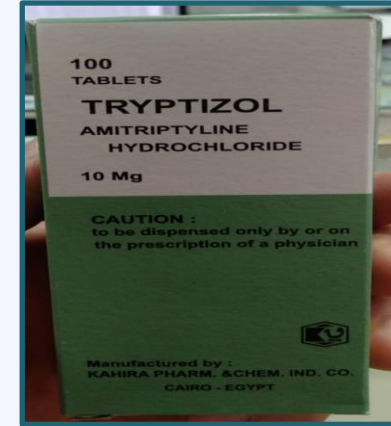
The following adverse effects are for MAO inhibitors only:

- Cheese Reaction (Hypertensive Crisis):
 - Certain foods like yogurt contain tyramine (has a sympathomimetic effect) which is metabolized by MAO.
 - Patients on MAO.I can not metabolize tyramine leading to its accumulation in the body
 - This accumulation results in:
 - Headache , Tachycardia ,hypertension , arrhythmia And seizures.
- MAO inhibitors are contraindicated in combination with other antidepressants to avoid Serotonin syndrome.
- CNS stimulation.



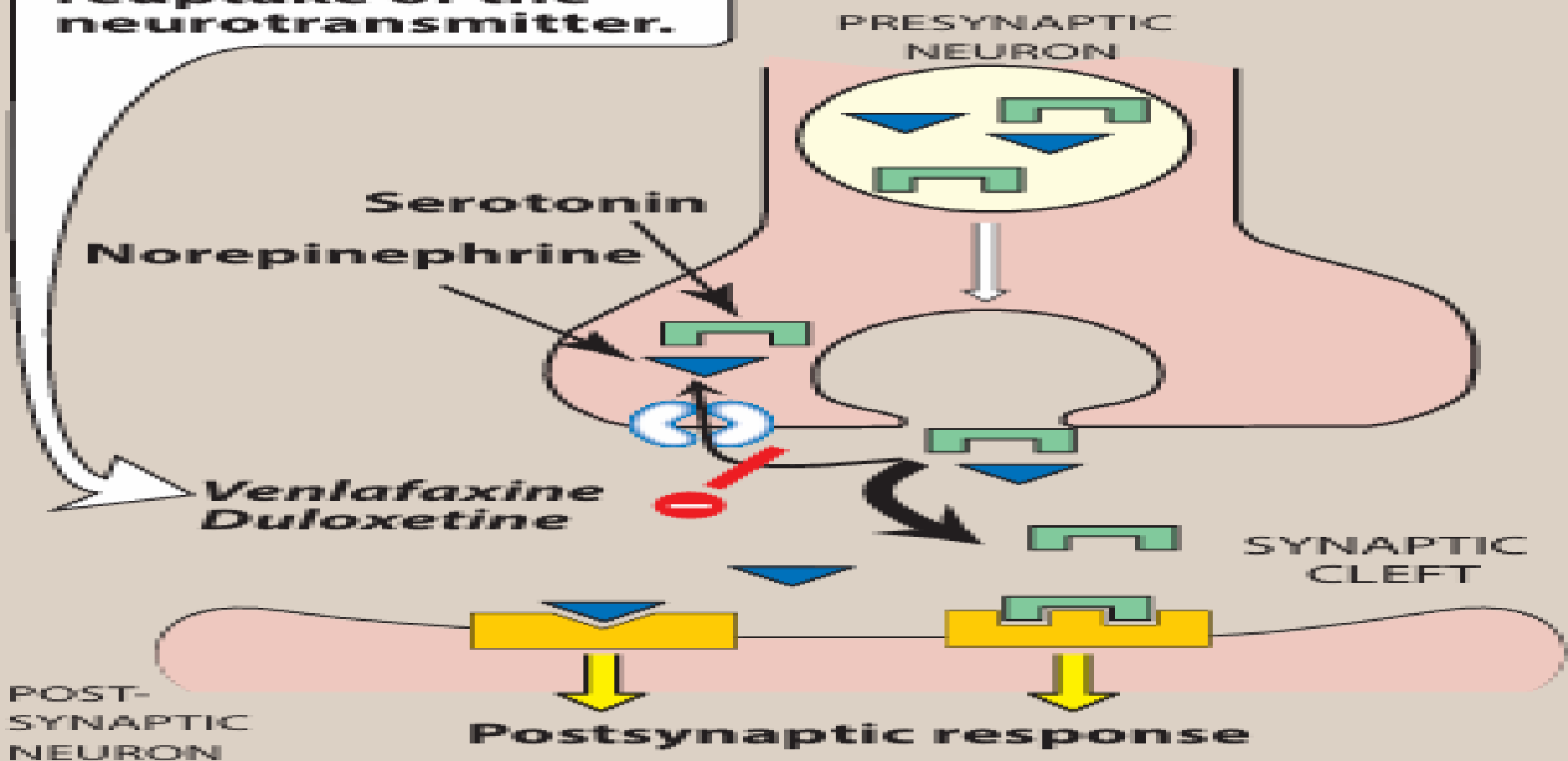
Tricyclic anti-depressants

- Members include :
 - Imipramine
 - Clomipramine
 - Amitriptyline
 - Nor-amitriptyline .



- MOA : Inhibition of Serotonin and Norepinephrine reuptake thus increasing the amount available in the synaptic space.
- Therapeutic Uses:
 - Major depression.
 - Chronic pain (Fibromyalgia)
 - Enuresis (Bed wetting problem) due to their blocking of muscarinic receptors.

Antidepressant drug blocks reuptake of the neurotransmitter.



- Adverse effects:

- CNS sedation (due to Atropine like action).
- Atropine like adverse effects.
- Hepatotoxicity in the form of cholestatic jaundice (very rare).

- (TCA toxicity):

- This toxicity is associated with large doses of the drug often suggesting a suicide attempt.
- Clinical manifestations are:
 - Atropine like toxicity.
 - Metabolic acidosis.
 - Cardiac arrhythmia.
- Treatment:
 - I.V NaHCO_3
 - I.V lidocaine if arrhythmia is not treated by NaHCO_3



+ Selective Serotonin Reuptake inhibitor

- Members include:

- Sertraline
- fluoxetine
- Citalopram
- escitalopram
- paroxetine.



- Therapeutic Uses:

- SSRI are first line treatment in major depression due to their safety even in large doses
- Obsessive compulsive disorder (OCD).
- Anxiety. (GAD)

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- Adverse effects:
 - CNS sedation or stimulation (usually disappears within 1-2 weeks).
 - GIT disturbance (prescribe PPI) and erectile dysfunctions are the most common side effects of SSRI and they disappear after 2 weeks of treatment.
 - SSRIs must be stopped gradually (over 4 weeks) to avoid Acute attack of Depression.
 - Weight gain or weight loss which is drug dependent.
 - Overdose with SSRIs does not usually cause cardiac arrhythmias, with the exception of citalopram, which may cause QT prolongation, however, QT prolongation with overdose is still a risk.
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Atypical Antidepressants:

- Noradrenaline and Serotonin reuptake inhibitors :
 - Venlafaxine and duloxetine.
 - Duloxetine is commonly prescribed for neuropathic pain.
- Receptor Blocker: mirtazapine and trazodone
 - Trazodone blocks 5HT_{2A} and 5HT_{2C}
 - Mirtazapine blocks 5HT_{2C}
 - Mirtazapine causes Less GIT upset and sexual dysfunction
 - And thus, can used in case if the patient wants to stop SSRI, first by gradual withdrawal of the SSRI then using mirtazapine.
- SNRI & Serotonin receptor blockers are used similarly to SSRI

Brands



Bupropion is a weak dopamine and norepinephrine reuptake inhibitor that is used to alleviate the symptoms of depression.

- Bupropion is also useful for decreasing cravings and attenuating withdrawal of cigarette smoking.
- Side effects may include dry mouth, sweating, nervousness, tremor, and a dose dependent increased **risk for seizures.**



+ Patient specific considerations in antidepressant drug selection

- Given that many antidepressants are equally efficacious despite having different mechanisms of action and adverse effects. It becomes clinically significant to choose the best antidepressant for a specific patient in an individualized process.
- There are two aspects which come into consideration when choosing the best antidepressant for a patient:
 - Primary aspects for the patient.
 - Secondary considerations.

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- Primary aspects for the patient:
 - Patient history with a specific antidepressant use.
 - Family history of antidepressant use
 - Adverse effect profile
 - Clinical presentation of depression
 - Comorbid medical/psychiatric conditions.
 - Secondary considerations include:
 - Age
 - Medication cost
 - Drug-food / drug-drug interactions
 - Genetic factors
 - Non-pharmacological intervention (psychotherapy)
 - History of non adherence to treatment.



Examples:

- Mirtazapine is known to cause weight gain and can cause poor compliance to treatment.
 - Bupropion is known to cause CNS stimulation and can be beneficial for patients experiencing hypersomnolence and detrimental to therapy when patient is experiencing insomnia.
 - The GIT and ED side-effects of SSRI often cause poor patient compliance and patient counseling is required.
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Sedating; may be useful for agitation

**SELECTIVE
SEROTONIN RE-UPTAKE
INHIBITORS**

Citalopram
Escitalopram
Fluoxetine
Fluvoxamine
Paroxetine
Sertraline

**SEROTONIN/
NOREPINEPHRINE
REUPTAKE INHIBITORS**

Duloxetine
Venlafaxine
Desvenlafaxine

**ATYPICAL
ANTIDEPRESSANTS**

Bupropion
Mirtazapine
Nefazodone
Trazodone

**TRICYCLIC/POLYCYCLIC
ANTIDEPRESSANTS**

Amitriptyline
Amoxapine
Clomipramine
Desipramine
Doxepin
Imipramine
Maprotiline
Nortriptyline
Protriptyline
Trimipramine

**MONOAMINE OXIDASE
INHIBITORS**

Phenelzine
Tranylcypromine

**High potential
for orthostatic
hypotension**

**Weight
gain**

**Gastro-
intestinal
distress**

Cases

- A 51-year-old woman with symptoms of major depression also has angle-closure glaucoma. Which of the following antidepressants should be avoided in this patient?
 - A. Amitriptyline.
 - B. Sertraline.
 - C. Bupropion.
 - D. Mirtazapine.
 - E. Fluvoxamine.

Answers

- Correct answer = A. Because of its potent antimuscarinic activity, amitriptyline should not be given to patients with glaucoma because of the risk of acute increases in intraocular pressure. The other antidepressants all lack antagonist activity at the muscarinic receptor.

Cases

- A 25-year-old woman has a long history of depressive symptoms accompanied by body aches and pain secondary to a car accident 2 years earlier. Physical and laboratory tests are unremarkable. Which of the following drugs might be useful in this patient?
 - A. Fluoxetine.
 - B. Sertraline.
 - C. Phenelzine.
 - D. Mirtazapine.
 - E. Duloxetine.

Answers

- Correct answer = E. Duloxetine is a SNRI that can be used for depression accompanied by symptoms of pain. SSRIs (fluoxetine and sertraline), MAOIs (phenelzine), and atypical antidepressants (mirtazapine) have little activity against pain syndromes.

Cases

- A 36-year-old man presents with symptoms of compulsive behavior. If anything is out of order, he feels that “work will not be accomplished effectively or efficiently.” He realizes that his behavior is interfering with his ability to accomplish his daily tasks but cannot seem to stop himself. Which of the following drugs would be most helpful to this patient?
 - A. Imipramine.
 - B. Fluvoxamine.
 - C. Amitriptyline.
 - D. Tranylcypromine.

Answers

- Correct answer = B. SSRIs are particularly effective in treating obsessive-compulsive disorder, and fluvoxamine is approved for this condition. The other drugs are less effective in the treatment of obsessive-compulsive disorder.

Cases

- A 55-year-old teacher began to experience changes in mood. He was losing interest in his work and lacked the desire to play his daily tennis match. He was preoccupied with feelings of guilt, worthlessness, and hopelessness. In addition to the psychiatric symptoms, the patient complained of muscle aches throughout his body. Physical and laboratory tests were unremarkable. After 6 weeks of therapy with fluoxetine, his symptoms resolved. However, the patient complains of stomach pains. Which of the following drugs might be useful in this patient?
 - A. Fluvoxamine.
 - B. Sertraline.
 - C. Citalopram.
 - D. Mirtazapine.

Answers

- D. Mirtazapine is largely free from gastric and sexual side effects, commonly associated with SSRIs (fluvoxamine, sertraline, and citalopram), as well as with TCAs, and SNRIs.

Discussions

- **1. The First-Line Controversy & The Pharmacist's Role**
- **Opening Question:** "The slides mention that psychotherapy, with or without medication, is the first-line treatment for depression, and that there's a debate about prescribing antidepressants alone. As the medication experts on the care team, what is our role when a prescriber wants to start an SSRI for a new patient?"
- **Potential follow-ups:**
 - "What key points should we include when counseling a patient starting their first antidepressant?"
 - "How do we manage patient expectations about the 2-week lag time for efficacy?"

Discussions

- **2. Adverse Effect Management & Patient Counseling**
- **Scenario:** "A patient on sertraline returns to the pharmacy after 10 days and complains of significant nausea and GIT upset. They are ready to stop the medication. What do we do?"
- **Potential follow-ups:**
 - How would your counseling for this side effects? (e.g., GIT upset)"
 - "The slides mention that mirtazapine can be a good alternative when SSRI side effects are intolerable. Why is that? What is its unique side effect profile we must warn about?"

Discussions

- **Case Study Rapid Fire & Rationale**
- **Use the cases from the lecture but ask for the clinical reasoning behind the answer.**
- **Case 1 (Glaucoma):** "Why exactly is amitriptyline contraindicated in angle-closure glaucoma? Let's talk about the receptor mechanism."
- **Case 2 (Chronic Pain):** "We chose duloxetine. Beyond 'it treats pain,' what is the pathophysiological reason an SNRI would be more effective for neuropathic pain than an SSRI?"

Discussions

- **Beyond the Pill: The Pharmacist's Holistic Role**
- **Discussion Starter:** "Let's look at the slide on 'Patient specific considerations.' A 22-year-old student and a 65-year-old retiree with the same diagnosis of major depression will likely have very different treatment plans. What factors from that list might be most important for each?"
- **Potential follow-ups:**
 - "For the student, how might medication cost and drug-drug interactions (e.g., with stimulants) influence our choice?"
 - "For the older adult, how do we factor in comorbidities and other medications (polypharmacy) when selecting an antidepressant?"
 - "How do we identify a patient who might be non-adherent and what strategies can we, as pharmacists, employ to help?"